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Education

2018-2020* **Ph.D.**, Ecology, L'Institut Agro Montpellier
2016-2017 **M.Sc.**, Computer Science, Ecole Supérieure d'Informatique - Alger
2012-2017 **B.Eng.** Computer Science, Ecole Supérieure d'Informatique - Alger

Research Experience

2022– Post-doctoral Fellow, [Laboratoire d'Ecologie Alpine \(LECA\)](#), Centre National de la Recherche Scientifique (CNRS), (Advisor: [Dr. Wilfried Thuiller](#))
2021-2021 Post-doctoral Fellow, [Biologie Computationnelle et Modélisation \(BCM\)](#) - TIMC, Université Grenoble Alpes, (Advisor: [Pr. Olivier François](#))
2018-2020 PhD candidate, [UMR Eco&Sols - Ecologie Fonctionnelle & Biogéochimie des Sols & des Agro-écosystèmes](#), Institut national de recherche pour l'agriculture, l'alimentation et l'environnement (INRAE) (Advisor: [Dr. Mickael Hedde](#))
2016-2017 Research Assistant, [LMCS - Laboratoire des Méthodes de Conception des Systèmes](#), Ecole Supérieure d'Informatique d'Alger (ESI) (Advisor: [Pr. Yacine Challal](#))

Research projects

2023-2026 **MODIC** - MOdélisation de la DIstrbution de la biodiversité et de la Connectivité des aires protégées
Role: Lead work package
Funding: LECA (CNRS) and PatriNat (OFB-MNHN) agreement
Coordinators: Florian Barnier (PatriNat)
2023-2025 **Natura Connect:** <https://naturaconnect.eu/>
Role: Co-lead work package
Funding: Horizon Europe Project
Coordinators: Piero Visconti (IIASA), Nestor Fernandez (iDiv)

*Expected.

2021–2023 **EO4DIVERSITY:** <https://www.eo4diversity.info/>
Role: Co-lead task
Funding: European Space Agency Biodiversity+ Precursor Programme
Coordinators: Andrew Skidmore (ITC, University of Twente), Margarita Huesca Martinez (ITC, University of Twente), Marc Paganini (ESA)

Publications

 [Google Scholar](#)

† → Equal contribution

Journal Articles

- . Hu, Y., Si-Moussi†, S. & Thuiller, W. Introduction to deep learning methods for multi-species predictions. *Methods in Ecology and Evolution* **16**, 228–246 (2025).
- . Bystrova, D., Assaad, C. K., Si-Moussi, S. & Thuiller, W. Causal discovery from ecological time-series with one timestamp and multiple observations. *bioRxiv*, 2024–10. <https://doi.org/10.1101/2024.10.10.608447> (2024).
- . Hartig, F., Abrego, N., Bush, A., Chase, J. M., Guillera-Arroita, G., Leibold, M. A., Ovaskainen, O., Pellissier, L., Pichler, M., Poggiato, G., *et al.* Novel community data in ecology-properties and prospects. *Trends in Ecology & Evolution* **39**, 280–293. <https://doi.org/10.1016/j.tree.2023.09.017> (2024).
- . Martinez-Almoyna, C., Calderòn-Sanou, I., Lionnet, C., Gielly, L., Boyer, F., Dufour, P., Dunyach, L., Miquel, C., Ohlmann, M., Poulenard, J., *et al.* Vegetation structure and climate shape mountain arthropod distributions across trophic levels. *Journal of Animal Ecology* **93**, 1510–1523. <https://doi.org/10.1111/1365-2656.14164> (2024).
- . Lamperti, L., Sanchez, T., Si Moussi, S., Mouillot, D., Albouy, C., Flück, B., Bruno, M., Valentini, A., Pellissier, L. & Manel, S. New deep learning-based methods for visualizing ecosystem properties using environmental DNA metabarcoding data. *Molecular Ecology Resources* **23**, 1946–1958. <https://doi.org/10.1111/1755-0998.13861> (2023).
- . Hedde, M., Blight, O., Briones, M. J., Bonfanti, J., Brauman, A., Brondani, M., Sanou, I. C., Clause, J., Conti, E., Cortet, J., *et al.* A common framework for developing robust soil fauna classifications. *Geoderma* **426**, 116073. <https://doi.org/10.1016/j.geoderma.2022.116073> (2022).
- . Bloor, J. M., Si-Moussi†, S., Taberlet, P., Carrère, P. & Hedde, M. Analysis of complex trophic networks reveals the signature of land-use intensification on soil communities in agroecosystems. *Scientific Reports* **11**, 18260. <https://doi.org/10.1038/s41598-021-97300-9> (2021).
- . Calderón-Sanou, I., Münkemüller, T., Zinger, L., Schimann, H., Yoccoz, N. G., Gielly, L., Foulquier, A., Hedde, M., Ohlmann, M., Roy, M., *et al.* Cascading effects of moth outbreaks on subarctic soil food webs. *Scientific Reports* **11**, 15054. <https://doi.org/10.1038/s41598-021-94227-z> (2021).

Peer-reviewed Conference Proceedings

- . Si-Moussi, S., Hedde, M. & Thuiller, W. *Plant recommendation using the environment and biotic associations* in *Working Notes of CLEF 2019 - Conference and Labs of the Evaluation Forum* (Lugano, Switzerland, Sept. 2019). <https://hal.science/hal-02970576>.

Preprints

- . Cerna, S., Si-Moussi, S., Calderon-Sanou, I., Miele, V. & Thuiller, W. *Limits and Promises of Earth Observation Foundation Models in Predicting Multi-Trophic Soil Biodiversity* ArXiv[Preprint]. 2025. <https://www.biorxiv.org/content/10.1101/2025.05.16.654504v1>.
- . Si-Moussi, S., Hennekens, S., Mucher, S., Los, S. & Thuiller, W. *Continental scale habitat modelling with artificial intelligence and multimodal earth observation* 2025. arXiv: [2507 . 09732](https://arxiv.org/abs/2507.09732) [cs.LG]. <https://arxiv.org/abs/2507.09732>.
- . Si-Moussi, S., Hennekens, S., Mûcher, S., De Keersmaecker, W., Chytrý, M., Agrillo, E., Attorre, F., Biurrun, I., Bonari, G., Čarni, A., et al. *EUNIS Habitat Maps: Enhancing Thematic and Spatial Resolution for Europe through Machine Learning* ArXiv[Preprint], Revision @Scientific Data. 2025. <https://arxiv.org/abs/2506.13649>.
- . Si-moussi, S., Thuiller, W., Galbrun, E., Decaëns, T., Gérard, S., Marchán, D. F., Marsden, C., Capowiez, Y. & Hedde, M. *Digging deeper: deep joint species distribution modeling reveals environmental drivers of Earthworm Communities* ArXiv[Preprint], Submitted to @Soil Biology & Biochemistry. 2025. <https://arxiv.org/abs/2506.13568>.
- . Si Moussi, S., Galbrun, E., Hedde, M., Poggiato, G., Rohr, M. & Thuiller, W. *Uncovering symmetric and asymmetric species associations from community and environmental data* working paper or preprint. Dec. 2022. <https://hal.science/hal-05161556>.

Datasets

- . Si-Moussi, S., Hennekens, S., Mucher, S. & Thuiller, W. *Current and future scenarios of EUNIS habitats distribution in Europe*. [Dataset v1](#). 2025.
- . Si-Moussi, S., Hennekens, S., Mucher, S. & Thuiller, W. *EUNIS Habitat Maps: Enhancing Thematic and Spatial Resolution for Europe through Machine Learning*. [Dataset v1](#). 2024.
- . Si-Moussi, S. & Thuiller, W. *Species habitat suitability of European terrestrial vertebrates for contemporary climate and land use*. [Dataset v1](#). 2024.

Presentations

Keynotes and Invited Seminars

- . Si-Moussi, S. & Cerna, S. *Deep modelling of ecological communities* DeepSDM conference. [Available on YouTube](#). Montpellier, France, June 2025.
- . Si-Moussi, S. & Thuiller, W. *Current and Future EUNIS Habitat Modeling and Mapping Biodiversa+* Working Group Habitat Pilot Seminar. Dec. 2024.
- . Thuiller, W. & Si-Moussi, S. *L'écologie prédictive à l'heure des données massives* TDMA — Traitement des Données Massives et Apprentissage. France, 2023.

- . Si-Moussi, S. *Building ecological networks: challenges and opportunities for Artificial Intelligence* Journées INRAE–Inria. Rennes, France, Nov. 2020.
- . Si-Moussi, S., Galbrun, E., Hedde, M. & Thuiller, W. *Machine Learning Ecological Interaction Networks* Journées du Groupe de Recherche en Écologie Trophique (GRET). Held at INRAE, 11–13 June 2019. Metz, France, June 2019.

Talks

- . Cerna, S., Si-Moussi, S., Miele, V. & Thuiller, W. *How do Earth Observation Foundation Models Help to Predict Multi-Trophic Soil Biodiversity* BioSpace25 - Biodiversity insight from Space. Frascati, Italy: European Space Agency (ESA), Feb. 2025.
- . Estopinan, J., Si-moussi, S., Giagnacovo, L., Miele, V. & Thuiller, W. *Using synthetic controls to attribute biodiversity shifts to remotely sensed landscape modifications* BioSpace25 - Biodiversity insight from Space. Frascati, Italy: European Space Agency (ESA), Feb. 2025.
- . Si-Moussi, S. *Introduction to Deep Learning Methods for Multi-Species Predictions*. Symposium Advances in Statistical Ecology - International Congress in Ecology & Evolution. Lyon, France, Oct. 2024.
- . Si-Moussi, S. *Harnessing AI and remote sensing for high resolution habitat mapping* 9th Annual Meeting - GdR Écologie Statistique. Lyon, France, May 2023.
- . Si-Moussi, S. & Hu, Y. *AI and Remote Sensing for Predictive Biodiversity Modelling* MIAI Meeting on AI & Ecology. Grenoble, France, June 2023.
- . Si-Moussi, S., François, O. & Thuiller, W. *Uncovering the Drivers of Soil Ecological Communities Turnover* Journée scientifique MSTIC–MIAI–PERSYVAL. Grenoble, France, Apr. 2022.
- . Si-Moussi, S. & Le Guillarme, N. *Applying Deep Learning Models to Ecological Data* MIAI Grenoble Alpes Meeting. Grenoble, France, Jan. 2022.
- . Thuiller, W., Si-Moussi, S., Calderon-Sanou, I., Le Guillarme, N. & Ohlmann, M. *A Machine-Learning Framework to Understand and Predict Soil Biodiversity* World Biodiversity Forum (WBF 2022). Davos, Switzerland, June 2022.
- . Si-Moussi, S. *Uncovering ecological associations from community and environmental data* Oral presentation at the 15th European Ecological Federation (EEF) Congress. Lisbon, Portugal, July 2019.
- . Si-Moussi, S. *A data-driven ecological interaction patterns discovery: case study of the soil fauna* Journées Jeunes Chercheurs 2018 - Société Française de Biométrie –. Paris, May 2018.
- . Si-Moussi, S. *Using trait-based joint species distribution models to understand the distribution of earthworms in France* TEBIS VII – Réseau TEBIS workshop. Avignon, France, Oct. 2018.

Posters

- . Si-moussi, S., Estopinan, J. & Thuiller, W. *Earth Observation foundation models for large-scale biodiversity modelling* BioSpace25 - Biodiversity insight from Space. Poster. European Space Agency (ESA), Feb. 2025.
- . Si-Moussi, S., Hennekens, S., Mûcher, F. & Thuiller, W. *Harnessing AI and Remote Sensing to Foster High Resolution Habitat Mapping* Poster. Davos, Switzerland, 2024.

Workshops

- Si-Moussi, S., Gibaud, J., Mortier, F. & Peyhardi, J. *Autour des modèles de distribution jointe d'espèces : les approches SCGLR, TRP et webSDM* Journées du GDR Ecologie Statistique. Montpellier, France, Nov. 2023.

Teaching

Department of Computer Science, University of Montpellier

2020	Teaching Assistant, Algorithmique & Complexité (HLIN401)
2019	Teaching Assistant, Introduction à l'Algorithmique et Programmation (HLIN101)

Faculty of Biology & Chemistry, Grenoble Alpes University

2022	Teaching Assistant, Modélisation en Biologie (BIO504)
2021	Teaching Assistant, Modélisation en Biologie (BIO504)

Academic Advising

Post-docs

2024-2025	Selene Cerna, Data Science, Earth Observation Foundation Models for biodiversity modelling
2022-2024	Yuqing Hu, Data Science, High resolution multi-modal models of Alpine plant communities

PhD steering committees

2025-	Baptiste Nicoud, Université Grenoble Alpes
2021-	Gabrielle Deschamps, Université Grenoble Alpes

Undergraduate

2024	Maxime Hoareau, ENS Lyon, Accounting for spatial constraints in species distribution models.
2022	Hichem Saghrouni, ENS Paris-Saclay, Modeling Invasive Alien Species (IAS) using remote sensing.
2021	Yoann Cartier, Grenoble INP Inse3, European Habitat Modeling using remote sensing and machine learning.
2019	Salah Eddine Bachir Cherif, ESI, Automated retrieval of ecological traits from scientific literature.
2018	Maeva Labouyrie, Montpellier SupAgro, Modelling soil food webs using eDNA.

Academic Service

Journal Reviewer

2021-2024 Methods in Ecology & Evolution

Awards & Honors

2019	GeoLifeCLEF Challenge (2nd place)
2018	Crop Data Challenge (1st place)
2017	INRA-INRIA PhD scholarship
2016	Study of the United States Institute (SUSI) scholarship